VARENDRA UNIVERSITY

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

ASSIGNEMENT-2 ON MAT 3141 (Applied Statistics & Probability)

Full Marks: 15

Instructions:

1. Solve all problems using any programming language such as Python/ R/ Java/ C++ or MATLAB
2. Programs must include proper input, processing, graphical visualization, and interpretation of results.
3. Submit source code, output screenshots, and brief explanations.

1. a) A machine learning research team collected data on training time (hours) and model accuracy (%) for different AI models. **4**

Model	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**
Training Time (hrs)	2	3	4	5	6	7	8	9
Accuracy (%)	60	65	68	72	78	82	88	91

i) Write a program to draw a scatter plot.
ii) Calculate the Karl's Pearson correlation coefficient.
iii) Interpret the strength and direction of the relationship.
iv) Display all results using programming output and graphs.

b) The following data represent GPU usage cost (in thousand TK.) and corresponding AI service revenue (in thousand TK.). **3.5**

GPU Cost	15	15	20	25	30	35
Revenue	25	35	45	50	60	72

i) Develop a linear regression model using programming.
ii) Plot the regression line with the dataset.
iii) Predict revenue when GPU cost = 28 thousand TK.
iv) Interpret the regression model with proper stat.

2. a) A software company recorded sprint completion times (days) for two development teams. **4**

Sprint	1	2	3	4	5	6
Team X	12	13	12	11	14	13
Team Y	8	10	12	15	18	20

i) Compare variability using QD, CQD, SD and CV.
ii) Determine which team is more consistent.
iii) Explain the importance of consistency in Agile development.
iv) Show program output and interpretation.

b) An IoT-based smart agriculture system recorded soil moisture sensor values. **3.5**

Reading	1	2	3	4	5	6	7	8	9	10
Moisture (%)	35	36	37	35	38	39	36	37	90	38

i) Draw a box plot using programming.

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ii) Detect outliers using the IQR method.
iii) Interpret the effect of abnormal readings on IoT decision systems.
iv) Display graphical and numerical outputs.